Mini Project Report On

Smart Meal Planner

Submitted in partial fulfillment of the requirements for the award of the degree of

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in

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By

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CERTIFICATE

This is to certify that the mini project report entitled ***"Smart Meal Planner"*** *is a bonafide record of the work done by* ***Manik Abrol (u2303216)***, ***George Mathew (u2303103)***, ***Diya Joby (u2303085)***, ***Densel Sabu (u2303081)***, *submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

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Abstract

The Smart Meal Planner System is developed to assist users in planning healthy and balanced meals according to their nutritional requirements and dietary preferences. In modern lifestyles, many individuals face difficulties in maintaining proper eating habits due to lack of time, knowledge about nutrition, and poor meal planning. This project aims to provide an intelligent and user-friendly system that helps users organize their daily meals in a structured and efficient manner.

The system allows users to input basic information such as calorie requirements, dietary preferences, and available food ingredients. Based on this information, the system generates appropriate meal suggestions that help maintain a balanced diet. The application is designed to simplify meal planning by automatically recommending suitable meals without requiring manual nutritional calculations.

Compared to traditional meal planning methods, the proposed system offers several advantages such as automated meal recommendations, improved nutritional awareness, and reduced time required for planning meals. It also provides a convenient platform where users can manage their daily food intake more effectively. By integrating technology with dietary planning, the system encourages healthier lifestyle habits and promotes better food management.

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Chapter 1

Database Tables

1.1 User Table

Field	Description
user_id	Unique ID of the user
username	Name of the user
password	User login password
age	Age of the user
gender	Gender of the user
height	Height of the user
weight	Weight of the user
activity_level	Physical activity level
body_goal	Fitness goal (lose/gain weight)

Table 1.1: Structure of User Table

1.2 Admin Table

Field	Description
admin_id	Unique admin ID
username	Admin username
password	Admin login password

Table 1.2: Structure of Admin Table

1.3 Meal Plan Table

Field	Description
plan_id	Unique meal plan ID
start_date	Plan starting date
end_date	Plan ending date
target_calories	Target calorie intake
user_id	Associated user ID

Table 1.3: Structure of Meal Plan Table

List of Abbreviations

SRS	Software Requirements Specification
UI	User Interface
DB	Database
SQL	Structured Query Language
API	Application Programming Interface
ER	Entity Relationship
BMR	Basal Metabolic Rate
BMI	Body Mass Index
TDEE	Total Daily Energy Expenditure
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
TDEE	Total Daily Energy Expenditure
AI	Artificial Intelligence
ML	Machine Learning
UX	User Experience
Kcal	Kilocalories

Chapter 2

Introduction

2.1 Background

Meal planning plays an important role in maintaining a healthy lifestyle and balanced nutrition. In today's fast paced society many individuals struggle to maintain proper dietary habits due to busy schedules limited nutritional knowledge and lack of structured meal planning methods. As a result unhealthy eating habits have become increasingly common which may lead to health problems such as obesity diabetes and other lifestyle related diseases.

With the advancement of technology digital solutions can help individuals manage their daily diet more efficiently. Software based systems can analyze nutritional data evaluate user preferences and recommend suitable meals based on dietary requirements. These systems reduce the effort required for manual planning and allow users to make better food choices.

The Smart Meal Planner system is developed as a solution to simplify the process of meal planning. The system assists users in organizing their meals according to their nutritional needs dietary preferences and available food ingredients. By providing structured meal recommendations the system helps individuals maintain a balanced diet and encourages healthier eating habits in everyday life.

2.2 Problem Definition

Many individuals face difficulty in maintaining a balanced diet due to lack of proper meal planning tools and nutritional awareness. The aim of this project is to develop a Smart Meal Planner system that assists users in generating meal plans based on their dietary preferences and nutritional requirements.

2.3 Scope and Motivation

The scope of the Smart Meal Planner project is to design and develop a system that assists users in organizing and planning their daily meals in an efficient manner. The system allows users to provide information such as dietary preferences, calorie requirements and available food ingredients. Based on this information the system analyzes nutritional data and suggests suitable meal options. The system focuses on simplifying the process of meal planning while helping users maintain balanced nutrition. It also provides a structured approach for managing daily food intake.

The motivation for this project arises from the increasing need for healthier eating habits in modern society. Many individuals do not have enough time or knowledge to plan nutritious meals regularly. As a result people often rely on unhealthy or unbalanced diets. By using technology to automate meal recommendations the Smart Meal Planner system aims to make diet planning easier and more accessible. The system encourages users to adopt healthier lifestyle choices and better dietary management.

Food Item	**Calories (kcal)**	**Category**
Apple	52	Fruit
Brown Rice	216	Grain
Grilled Chicken	165	Protein
Broccoli	34	Vegetable

Table 2.1: Sample Nutritional Data Used in the Smart Meal Planner

2.4 Objectives

- To design and develop a Smart Meal Planner system that assists users in planning balanced meals.
- To provide meal recommendations based on nutritional requirements and dietary preferences.
- To simplify the process of daily meal planning using a digital platform.

- To help users maintain a healthy lifestyle by encouraging proper dietary habits.
- To organize food items and nutritional information in a structured database.
- To develop a user-friendly interface that allows users to easily interact with the system.

2.5 Challenges

One of the major challenges in developing the Smart Meal Planner system is collecting accurate and reliable nutritional data for different food items. Another challenge is designing a system that can generate appropriate meal recommendations based on user preferences and dietary requirements. Ensuring that the system remains simple and easy to use while still providing useful and meaningful meal suggestions is also an important challenge.

2.6 Assumptions

TIt is assumed that users will provide accurate information regarding their dietary preferences and nutritional requirements while using the system. It is also assumed that the nutritional dataset used in the system contains correct and reliable information about food items. Another assumption is that users will access the system through a simple user interface to obtain meal recommendations.

2.7 Societal / Industrial Relevance

The Smart Meal Planner system has significant relevance for both society and industry as it helps individuals maintain healthier eating habits and better diet management. The system can assist people who wish to follow balanced diets but lack the time or knowledge required for proper meal planning. By providing structured meal recommendations, the system encourages users to adopt healthier lifestyles and improve their overall well-being.

In industrial and professional settings, the system can be useful in healthcare, fitness, and nutrition related industries where diet planning plays an important role. Nutritionists, fitness trainers, and healthcare professionals can use such systems to guide individuals in maintaining proper dietary routines. The application of this system can also support

digital health platforms and wellness applications that focus on improving public health and nutritional awareness.

2.8 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Smart Meal Planner system. Chapter 1 presents the introduction of the project including the background, problem definition, objectives, scope, and motivation of the work. Chapter 2 describes the Software Requirements Specification which outlines the functional and non-functional requirements of the system.

Chapter 3 explains the system architecture and design, including the overall structure of the system, dataset used, database design, and implementation strategies. The remaining sections provide supporting information such as references and additional materials related to the project.

Chapter 3

Software Requirements Specification

3.1 Introduction

Figure 3.1: Smart Meal Planner System Overview

3.2 Overall Description

The Smart Meal Planner system is designed to help users plan their meals based on their nutritional requirements and personal health goals. The system collects information such as age, weight, height, activity level, and fitness goals from the user. Based on this data, it generates a personalized meal plan that helps users maintain a balanced diet.

The application provides a simple and user-friendly interface where users can view recommended meals and their nutritional information. The system stores user details and food data in a database and uses this information to create weekly meal plans. The main objective of the Smart Meal Planner is to assist users in maintaining a healthy lifestyle by providing structured and customized meal recommendations.

3.3 External Interface Requirements

The system interacts with users through a graphical user interface where users provide their dietary preferences, calorie requirements and available ingredients. The system processes this input and generates suitable meal recommendations. Nutritional calculations are performed using basic formulas that estimate the total calorie intake from different food components.

$$Total\ Calories = Calories_{protein} + Calories_{carbohydrates} + Calories_{fat} \tag{3.1}$$

$$Calories_{protein} = Protein\ (g) \times 4 \tag{3.2}$$

$$Calories_{carbohydrates} = Carbohydrates\ (g) \times 4 \tag{3.3}$$

$$Calories_{fat} = Fat\ (g) \times 9 \tag{3.4}$$

3.4 System Features

The Smart Meal Planner system provides several features that assist users in planning balanced meals according to their dietary requirements. The system allows users to enter basic information such as dietary preferences, calorie requirements, and available food ingredients. Based on this information, the system analyzes nutritional data and generates appropriate meal recommendations.

The system organizes food items and their nutritional values in a structured database which enables efficient retrieval of information when generating meal plans. The application also presents meal suggestions in a clear and user friendly interface so that users can easily understand and follow the recommended diet.

In addition, the system helps users maintain a balanced intake of nutrients by suggesting meals that meet their calorie and dietary needs. By simplifying the process of meal planning, the system supports users in making healthier food choices and managing their daily diet effectively.

3.5 Other Nonfunctional Requirements

The Smart Meal Planner system should be reliable and capable of providing accurate meal recommendations based on the user's dietary preferences and nutritional require-

ments. The system should also be efficient so that users can receive meal suggestions quickly without delays.

The application should have a simple and user friendly interface that allows users to easily enter their preferences and understand the recommended meal plans. In addition, the system should ensure the privacy and security of user information. The software should also be maintainable and scalable so that new food items and nutritional data can be added in the future without affecting the overall performance of the system.

Chapter 4

System Architecture and Design

4.1 System Overview

The Smart Meal Planner system is designed to assist users in planning healthy and balanced meals based on their nutritional requirements and dietary preferences. The system provides a digital platform where users can input their personal dietary needs such as calorie requirements, preferred food types and available ingredients. Based on this information the system analyzes stored nutritional data and generates appropriate meal recommendations.

The overall working of the system involves several stages. Initially the user provides basic inputs through the user interface. These inputs may include dietary preferences, calorie limits, and specific food restrictions. Once the data is provided the system processes the information using predefined logic and compares it with the nutritional dataset available in the database.

The database contains information about different food items including their nutritional values such as calories, carbohydrates, proteins and fats. The system evaluates the input requirements and selects suitable food combinations that meet the nutritional needs of the user. These results are then presented to the user in the form of recommended meals.

The architecture of the Smart Meal Planner system consists of several main components including the user interface, data processing module, recommendation module and the nutritional database. The user interface allows users to interact with the system and provide necessary inputs. The data processing module analyzes the input data and applies the required calculations. The recommendation module generates suitable meal suggestions based on the processed information. Finally the database stores and manages the nutritional information of different food items used by the system.

By combining these components the Smart Meal Planner system provides an organized and efficient approach for meal planning. The system simplifies the process of selecting healthy meals and helps users maintain balanced nutrition in their daily lives.

4.2 Architectural Design

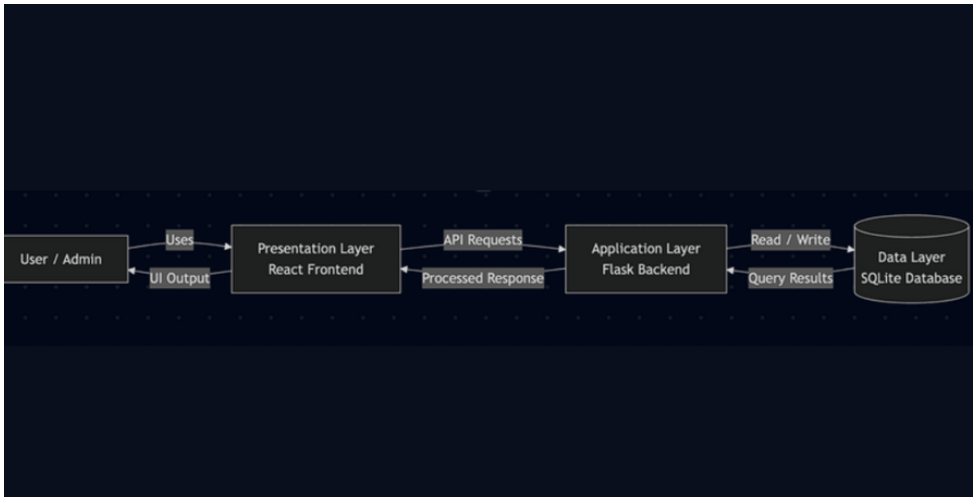

Figure 4.1: Sequence Diagram of Smart Meal Planner System

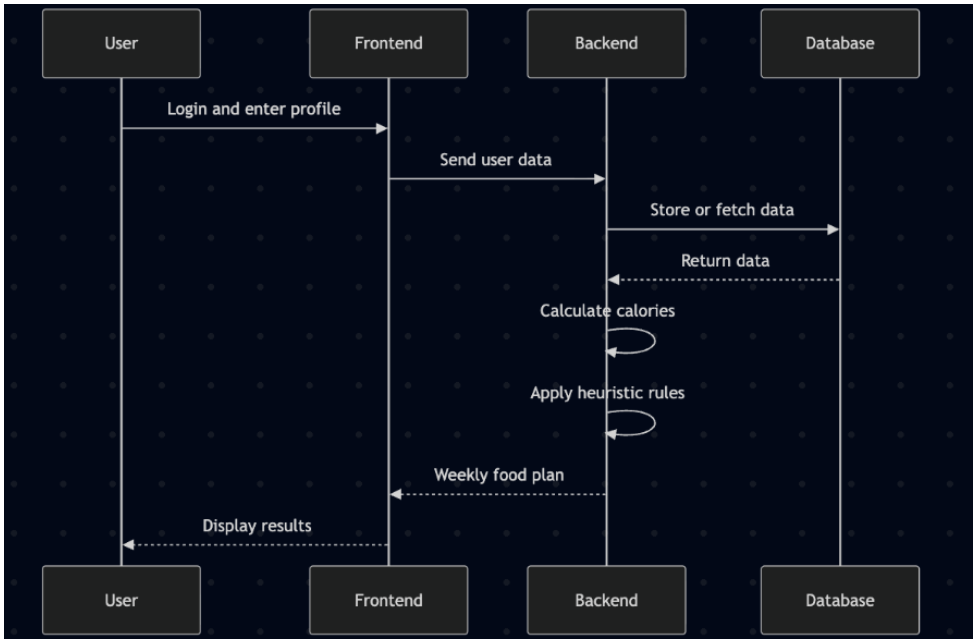

Figure 4.2: Entity Relationship Diagram of Smart Meal Planner System

4.3 Dataset identified

The Smart Meal Planner system uses a nutritional dataset that contains information about different food items and their nutritional values. The dataset includes attributes such as food name, calorie content, protein content, carbohydrate content, fat content and food category. This information helps the system evaluate the nutritional composition of different meals and generate appropriate meal recommendations based on user requirements.

The data used in the system is collected from publicly available nutritional databases and food nutrition resources. These datasets provide reliable information about common food items and their nutritional properties. The dataset is organized in a structured format so that the system can efficiently retrieve and process the required information during meal generation.

A sample subset of the dataset includes food items such as fruits, vegetables, grains, dairy products and protein sources along with their respective nutritional values. The system uses this dataset to compare the user's dietary requirements with the available food data and generate balanced meal suggestions that meet the specified calorie and nutritional limits.

4.4 Proposed Methodology/Algorithms

The Smart Meal Planner system follows a structured approach to generate meal recommendations based on user dietary requirements. The methodology begins with collecting user input such as dietary preferences, calorie requirements and available food ingredients. The system then processes this information and compares it with the nutritional dataset stored in the database. Based on this comparison the system selects suitable food combinations that satisfy the nutritional requirements of the user.

The algorithm analyzes the nutritional values of different food items such as calories, carbohydrates, proteins and fats. It ensures that the selected meals maintain a balanced distribution of nutrients and remain within the user's specified calorie limit. The final recommended meals are then displayed through the user interface.

Algorithm: Smart Meal Recommendation

Step 1: Start the system.

Step 2: Accept user input such as dietary preferences, calorie requirement and available ingredients.

Step 3: Retrieve food items and nutritional data from the database.

Step 4: Analyze the nutritional values of each food item.

Step 5: Compare user requirements with the nutritional dataset.

Step 6: Select suitable food combinations that satisfy the calorie and nutrient constraints.

Step 7: Generate recommended meal plans.

Step 8: Display the recommended meals to the user.

Step 9: End the process.

4.5 User Interface Design

The user interface of the Smart Meal Planner system is designed to provide a simple and user-friendly interaction for users. The interface allows users to enter their personal health information, view meal recommendations, and access their weekly meal plan.

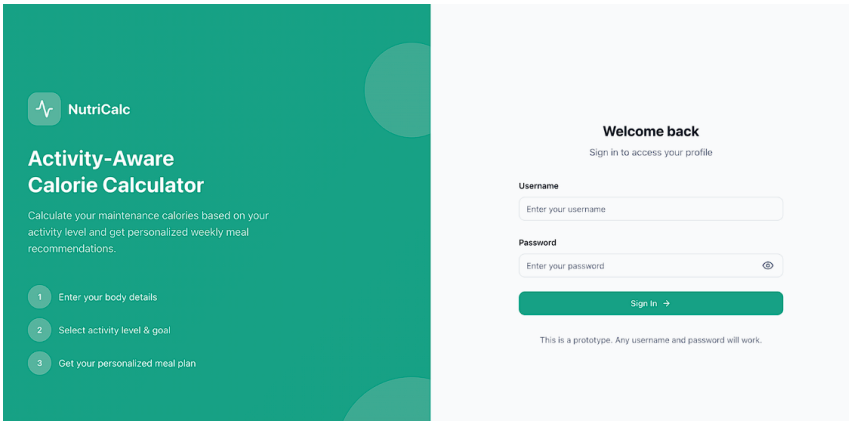

Figure 4.3: Login Interface of the Smart Meal Planner System

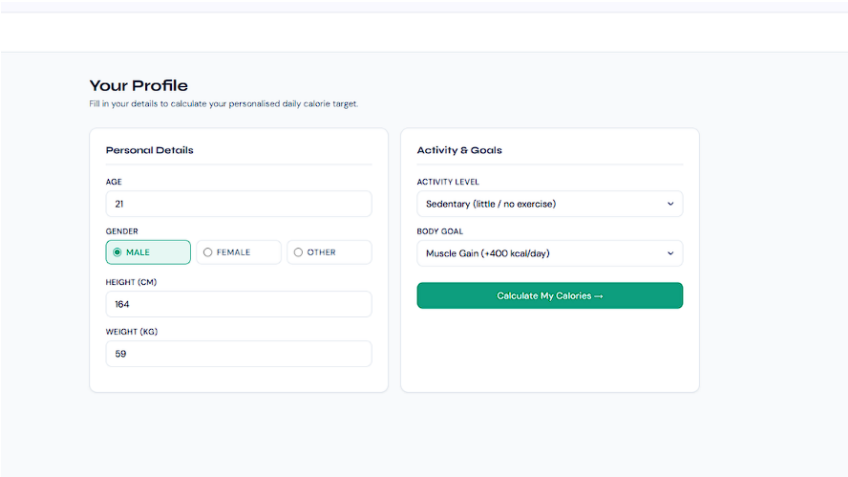

Figure 4.4: User Health Information Interface

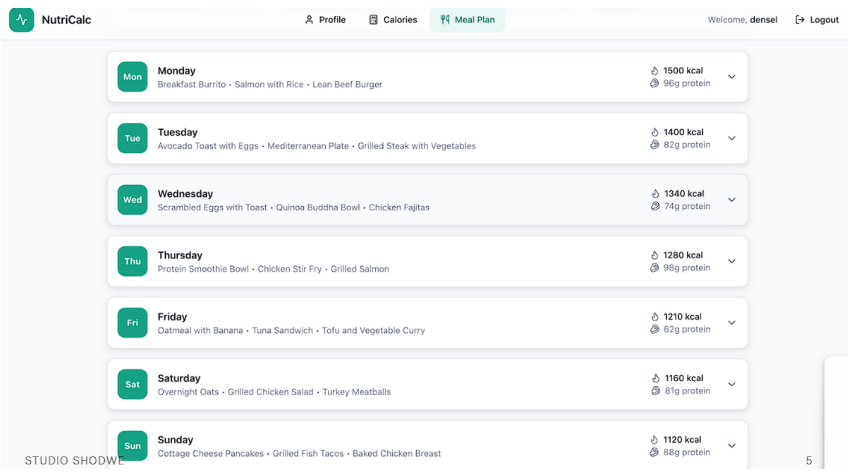

Figure 4.5: Meal Recommendation Interface

4.6 Database Design

The Smart Meal Planner system uses a relational database to store and manage information related to users, food items, and meal plans. The database includes tables such as User, Admin, Food, WeeklyPlan, and PlanMeal.

The User table stores personal and health details of users. The Food table contains information about food items including calories and protein. The WeeklyPlan table manages the meal plan created for each user, while the PlanMeal table links specific food items to the meal plan for different days and meal times. These tables are connected using primary and foreign keys to maintain proper relationships between the data.

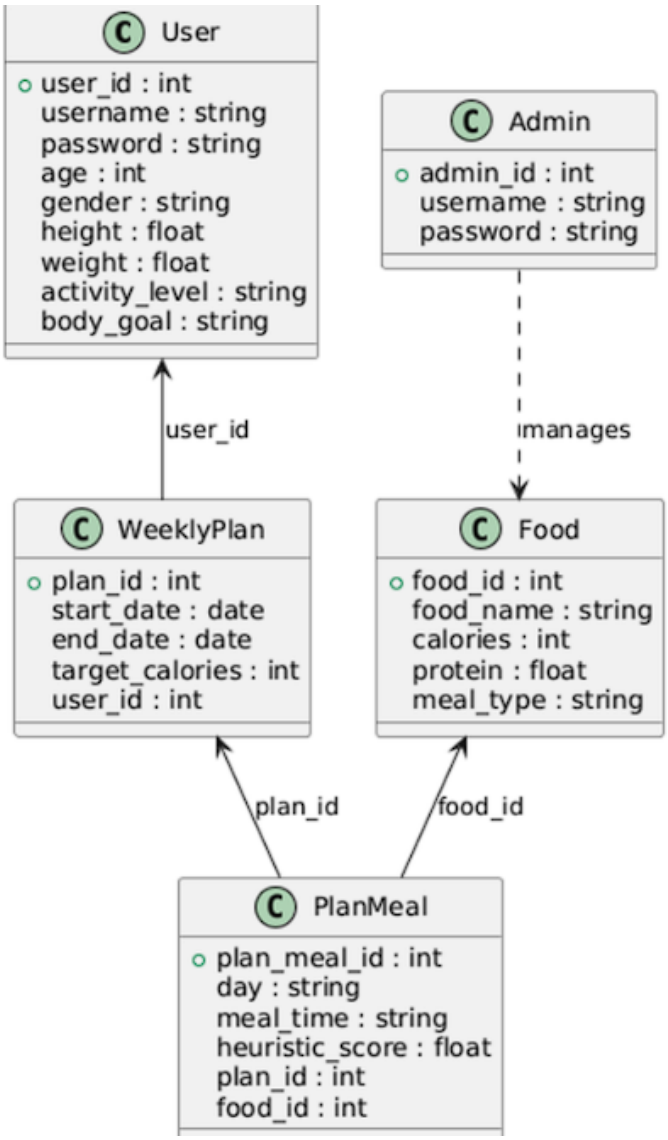

Figure 4.6: Database Schema of the Smart Meal Planner System

4.7 Description of Implementation Strategies

This section details the important implementation strategies used in your project. For example, (1) if you are doing a audio/ video capture, what is your python library and associated methods for doing that (2) if you are designing a CNN, how did you do that in the language which you are using for implementation (3) what are the different methods in the language you will use to evaluate your work. Small snippets of code can be supplemented to support your strategy.

4.8 Module Division

This project is divided into several modules to ensure efficient development and proper organization of tasks. Each team member was responsible for a specific part of the system.

Frontend Development and Backend Integration: This module focuses on designing the user interface and ensuring smooth interaction between the frontend and backend components of the system. This part was developed by Manik Abrol, who handled the frontend design and also established the connection between the frontend and backend.

Backend Development: The backend module is responsible for handling the server-side logic, processing user requests, and managing application functionality. This module was developed by Densel Sabu.

Database Management (SQL): This module involves designing and managing the database, including creating tables, storing data, and handling database queries using SQL. This part was implemented by George Mathew.

System Integration and Support Tasks: This module includes assisting with system organization, documentation, and supporting the overall development process. These tasks were handled by Diya Joby.

4.9 Work Schedule - Gantt Chart

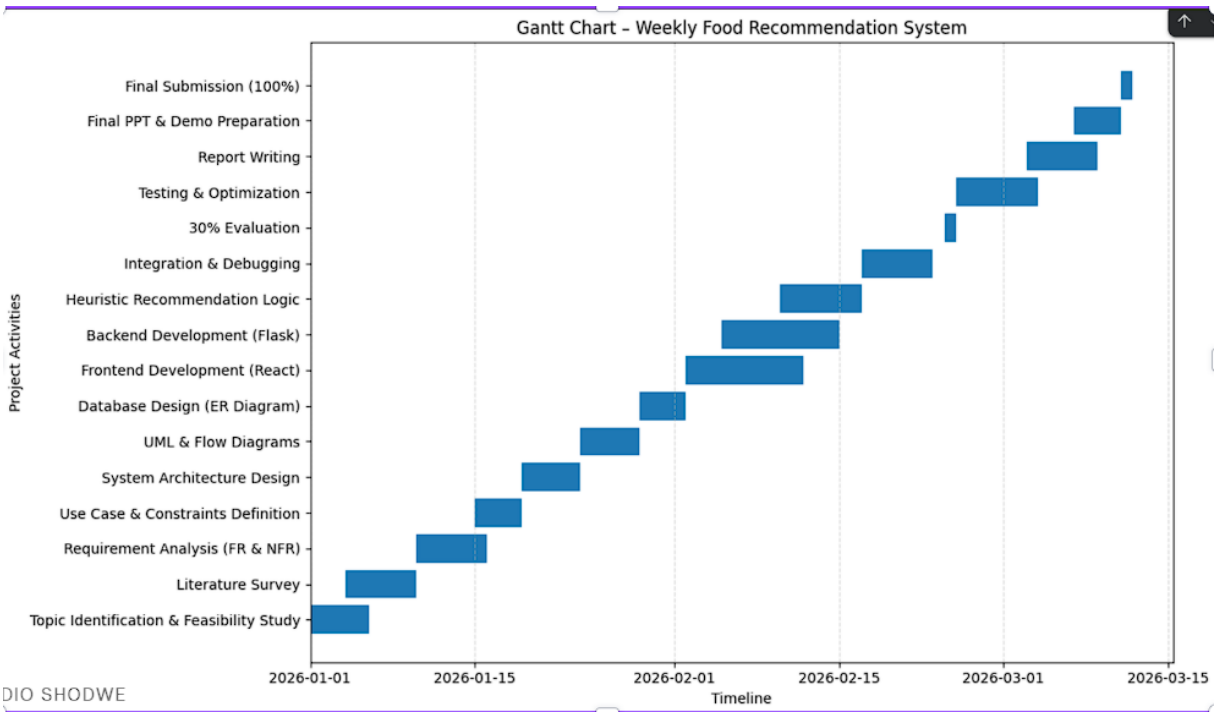

Figure 4.7: Gantt Chart of the Smart Meal Planner Project

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